

$$\begin{array}{ccc}
 (A \otimes \mathbf{1}) \otimes B & \xrightarrow{\alpha_{A, \mathbf{1}, B}} & A \otimes (\mathbf{1} \otimes B) \\
 & \searrow r_A \otimes \mathrm{id}_B & \swarrow \mathrm{id}_A \otimes l_B \\
 & A \otimes B &
 \end{array}$$